Warm-Up Answer Key

Expressions and Operations

LESSON 6-1 · DIVISION EXPRESSIONS WITH FRACTIONS AND WHOLE NUMBERS

MOVE *Name the reciprocal, then multiply / Read a fraction bar as "divided by"*
SLIDE 1) 2.4 2) 1/5 3) 2 4) 4
MODEL 4 4 12

1. 8	**5.** 5
2. 15	**6.** 3
3. 24	**7.** 4
4. 16	**8.** 6 pieces

DROPPED *skill 1 – dividing decimals belongs to the Unit 2 decimal thread; nothing in fraction division uses it*

LESSON 6-2 · DIVISION EXPRESSIONS WITH FRACTIONS AND MIXED NUMBERS

MOVE *Rewrite a mixed number as an improper fraction / Divide fractions: multiply by the reciprocal, then simplify*
SLIDE 1) 11/4 2) 2/15 3) 5/3 4) 3/4
MODEL 5/2 4/3 3 1/3

1. 6	**5.** 17/5
2. 4 2/3	**6.** 2 1/2
3. 1 1/8	**7.** 1/2
4. 1/2	**8.** 6 batches

LESSON 6-3 · EXPLORE NUMERICAL EXPRESSIONS WITH EXPONENTS

MOVE *Multiply a repeated factor to find its value*
SLIDE 1) 8 2) 36 3) 1,000 4) 4
MODEL 4 8 16

1. 8	**5.** 100
2. 27	**6.** 1,000
3. 25	**7.** 6
4. 64	**8.** 3 times, and 125

DROPPED *skill 4 – counting the written factors is how this lesson introduces the exponent itself, so it is the lesson's own move, not a prerequisite*

LESSON 6-4 · WRITE AND EVALUATE NUMERICAL EXPRESSIONS WITH EXPONENTS

MOVE *Evaluate with the order of operations, powers included*
SLIDE 1) 9 2) 23 3) 4 4) 4 × 7
MODEL 4 20 23

1. 20	**5.** 4
2. 7	**6.** 30
3. 25	**7.** 5
4. 2	**8.** 3 × 5 + 4 = 19

DROPPED *skill 4 – turning "sum," "product," and "less than" into symbols is session 1 of this lesson; item 8 bridges into it instead*

LESSON 6-5 · WRITE AND EVALUATE ALGEBRAIC EXPRESSIONS

MOVE *Substitute the value, then evaluate*
SLIDE 1) 14 2) 23 3) n 4) n + 7
MODEL 5 20 23

1. 20	**5.** 17
2. 11	**6.** 16
3. 23	**7.** 5
4. 12	**8.** n + 7

DROPPED *skill 4 — writing a word phrase as an algebraic expression is session 1 of this lesson; item 8 bridges into it instead*

LESSON 6-6 · IDENTIFY EQUIVALENT ALGEBRAIC EXPRESSIONS

MOVE *Evaluate two expressions at the same value and compare / Multiply a number by a sum*
SLIDE 1) 7x 2) 14 3) 27 4) yes, both are 14
MODEL 7 8 14

1. 21	**5.** 27
2. 16	**6.** 30 and 30
3. 28	**7.** 4
4. 20	**8.** 7x

DROPPED *skill 1 — combining like terms is what session 1 of this lesson teaches; item 8 bridges into it instead*

LESSON 6-7 · FIND FACTORS AND MULTIPLES

MOVE *List all the factors of a number, in pairs / List the first multiples of a number*
SLIDE 1) 56 2) 1, 2, 3, 6, 9, 18 3) 6, 12, 18, 24, 30 4) yes
MODEL 9 6 6

1. 1, 2, 3, 4, 6, 12	**5.** 6, 12, 18, 24, 30
2. 1, 2, 4, 5, 10, 20	**6.** 8, 16, 24, 32, 40
3. 1, 2, 3, 4, 6, 8, 12, 24	**7.** 15
4. yes	**8.** 6

LESSON 6-8 · GENERATE EQUIVALENT EXPRESSIONS

MOVE *Expand: multiply a number by a sum / Find the GCF of two numbers*
SLIDE 1) 6 2) 3x + 15 3) 7x 4) 42
MODEL 5 3x 15

1. 2x + 8	**5.** 6
2. 5x + 15	**6.** 10
3. 4n + 24	**7.** 7
4. 6x + 3	**8.** x + 3

DROPPED *skill 3 — combining like terms is rehearsed in the 6-6 warm-up and here it is only a step inside the lesson's own rewriting work, not a separate prerequisite move*